

Kognitionspsychologie II: Session 3

Neural basis of emotions

Loreen Tisdall, FS 2026

Version: March 10, 2026

Semester overview



#	Date	Topic	Slides	Instructor
1	17.02.2026	Emotion: What is an emotion?	pdf	Mata
2	03.03.2026	Emotion: What is an emotion? (continued)	pdf	Mata
3	10.03.2026	Emotion: Neural bases		Tisdall
4	17.03.2026	Emotion: Regulation		Mata
5	24.03.2026	Emotion: Well-being		Mata
6	31.03.2026	Motivation: What is motivation?		Mata
7	07.04.2026	Motivation: Extrinsic vs intrinsic motivation		Mata
8	14.04.2026	Motivation: Neural bases		Tisdall
9	28.04.2026	Motivation: Cooperation		Theisen
10	05.05.2026	Motivation: Morality		Theisen
11	12.05.2026	Applications		Mata
12	19.05.2026	Wrap-up and Q&A		Mata
13	03.06.2026	Exam (DSBG Neubau)		
14	20.07.2026	Repeat Exam (Vesalianum)		

Learning Objectives

- Learn about **neural model**(s) of basic (and not so basic) emotions
- Learn about methods to **meta-analyze** neuroimaging studies, and their shortcomings (and possible solutions ;))
- Engage with the evidence supporting **locationist versus constructivist approaches** to the neural basis of emotions
- Get a first intuition of how **brain maturation** supports the development of emotion control / regulation from childhood to early adulthood

Recap: What is emotion?

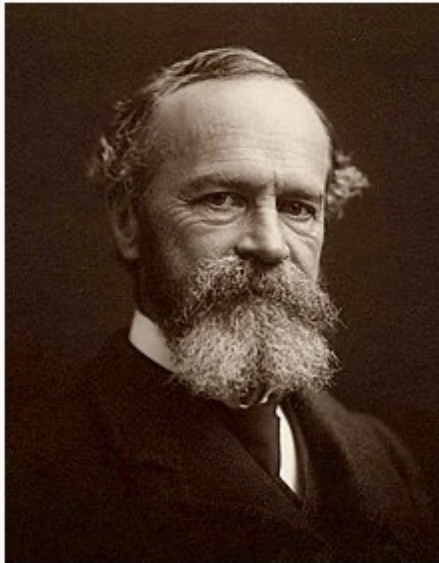


Image created with AI (Bing), January 31, 2024

“It is widely agreed that emotion refers to a collection of psychological states that include **subjective experience**, **expressive behavior** (e.g., facial, bodily, verbal), and **peripheral physiological responses** (e.g., heart rate, respiration). It is also widely agreed that emotions are a **central feature in any psychological model** of the human mind. **Beyond these two points of agreement, however, almost everything else seems to be subject to debate.**”

The neural basis of emotion

William James



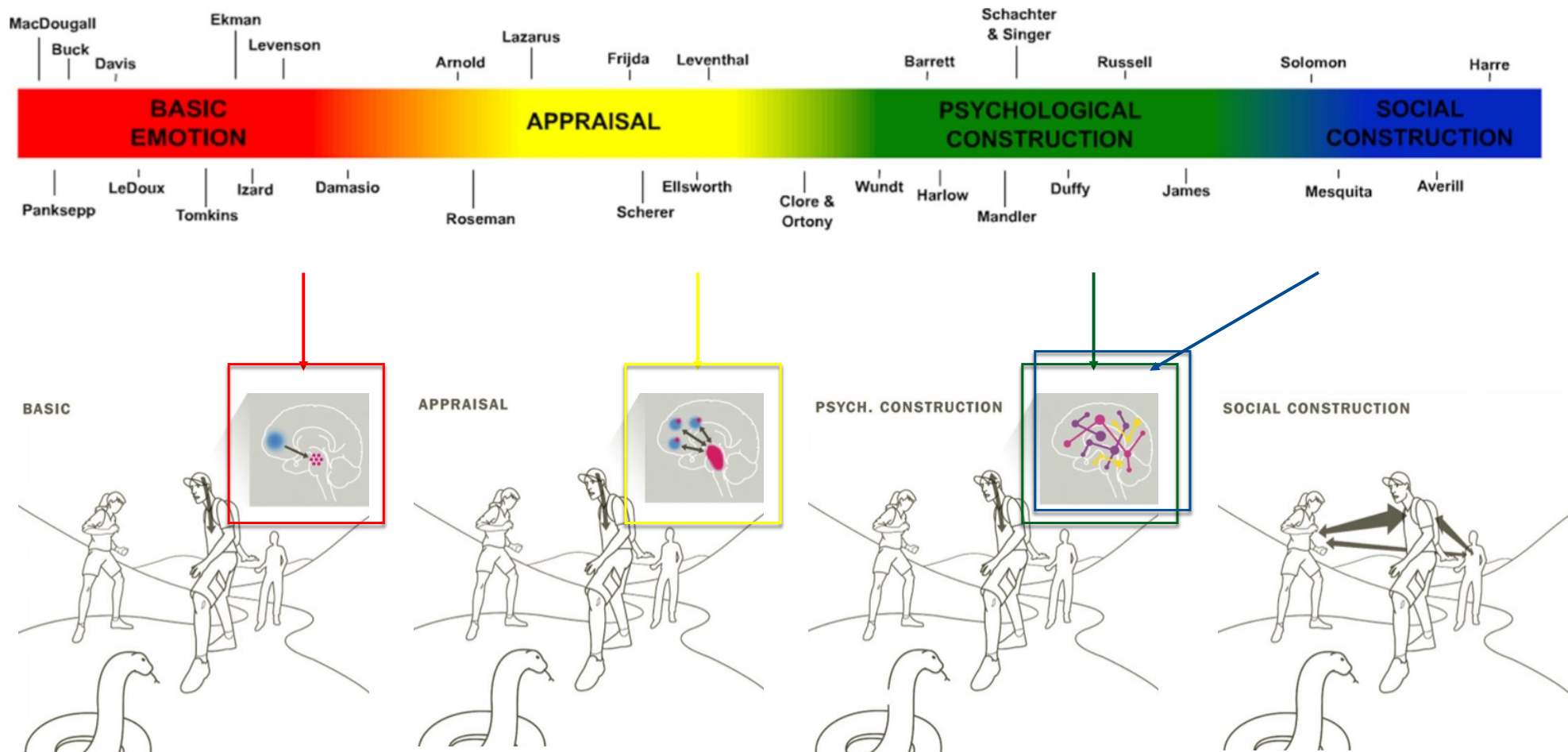
James in 1903

https://en.wikipedia.org/wiki/William_James

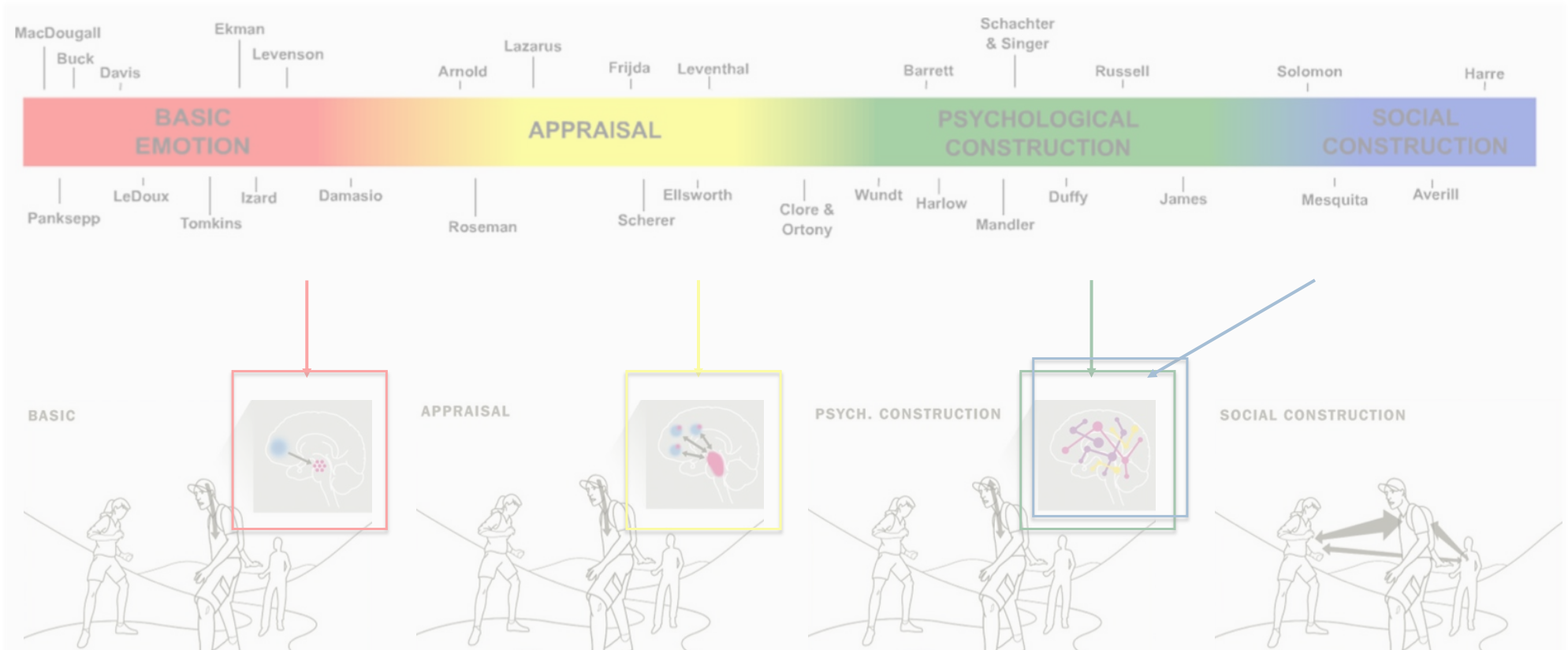
“ [...] of two things concerning the emotions, one must be true. Either separate and special centres, affected to them alone, are their brain-seat, or else they correspond to processes occurring in the motor and sensory centres already assigned.”

(William James, 1890, in Lindquist et al., 2012)

Recap: Four perspectives on emotion



Recap: Four perspectives on emotion



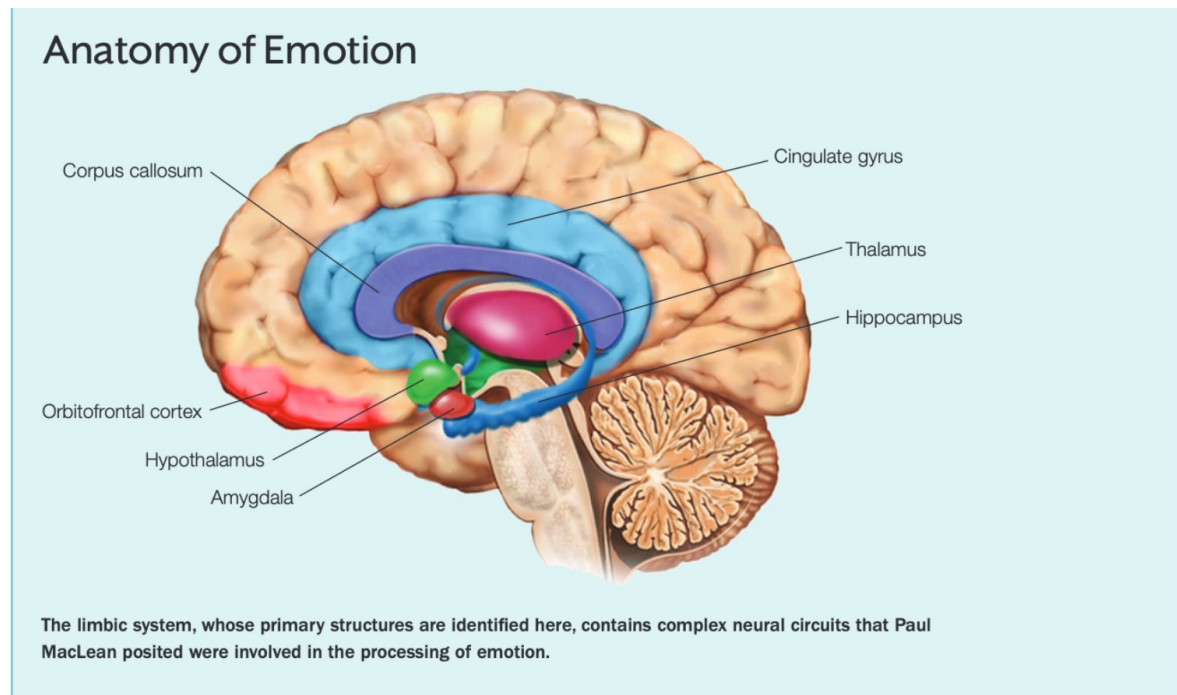
In affective neuroscience, this is framed as the “locationist vs. psychological constructionist” debate

Anatomical orientation: The "emotional" brain

Core structures of the limbic system

(NOTE: contentious!)

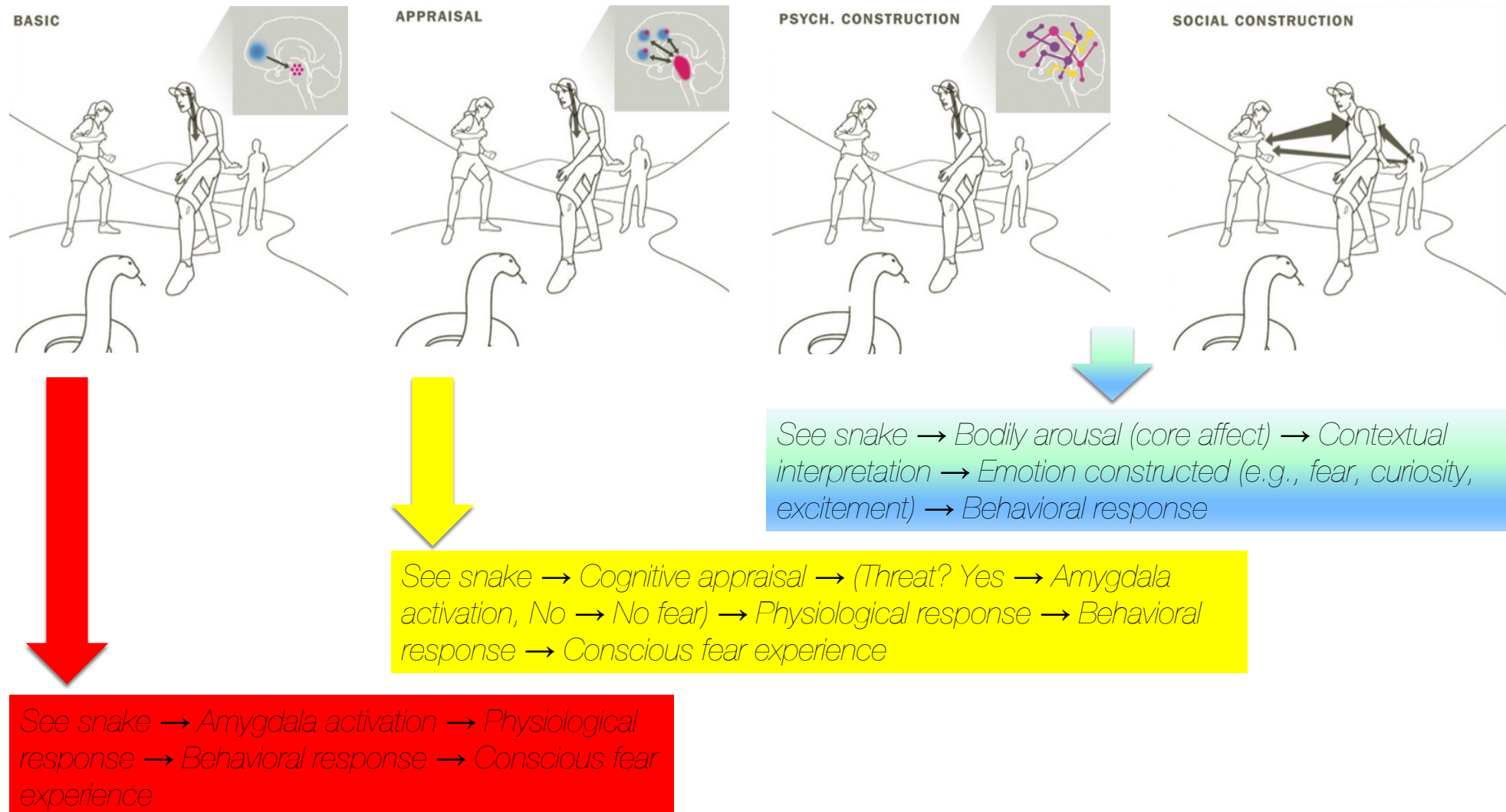
- Limbic lobe
- **Thalamus**
- **Hypothalamus**
- **Amygdala**
- **Hippocampus**
- OFC
- Basal ganglia



Locationist

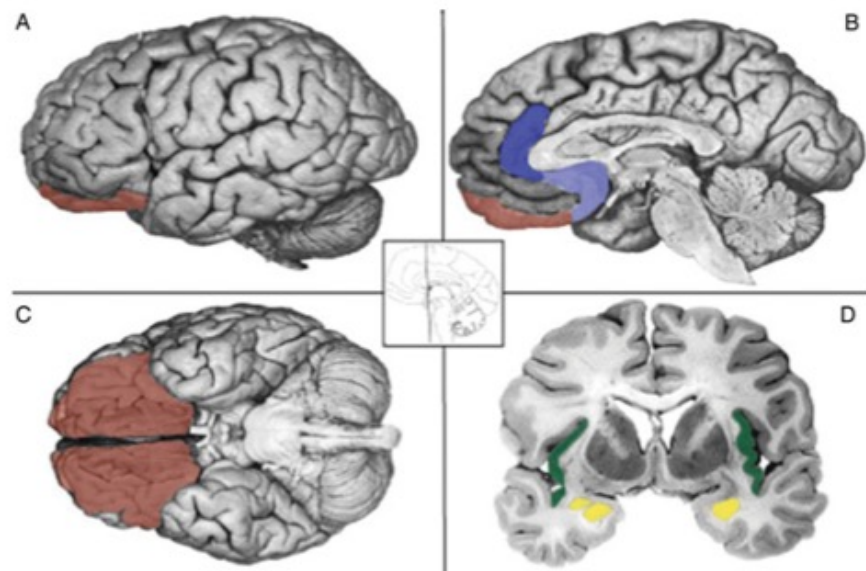
	Locationist → <i>natural kind / modal model of emotions</i>	Psychological constructionist → <i>conceptual act model</i>
Core assumptions	• The body and brain respect the category emotion and individual categories within • Emotion categories refer to states with endowed motivational characteristics that drive cognition and behavior • States are biologically basic and cannot be broken down further	• The body and brain do not respect emotion categories • Emotions are “situated conceptualizations”, that is, psychological events that emerge out of more basic psychological operations that are not specific to emotion • Emerge when people make meaning out of sensory input (e.g., from the body) using their experience, expectations, etc. • Emotion = “core affect” (mental representation of bodily changes) + “conceptualization” (using experience to make meaning out of current sensation) + “emotion words” + “executive attention”
Implications for the neural basis	• Emotional experience is rooted in dedicated brain mechanisms • Each emotion is supported by dedicated, innate neural circuits: discrete emotion categories consistently and specifically correspond to single distinct (architecturally defined) brain regions	• Discrete emotion categories are constructed from more general brain networks (“psychological primitives”) which are not specific to emotion • Categorization (or conceptualization) is a fundamental process in the human brain • Realized in a set of brain regions that reconstitutes prior experiences for use in the present (“episodic memory network”)

Locationist vs. constructionist account



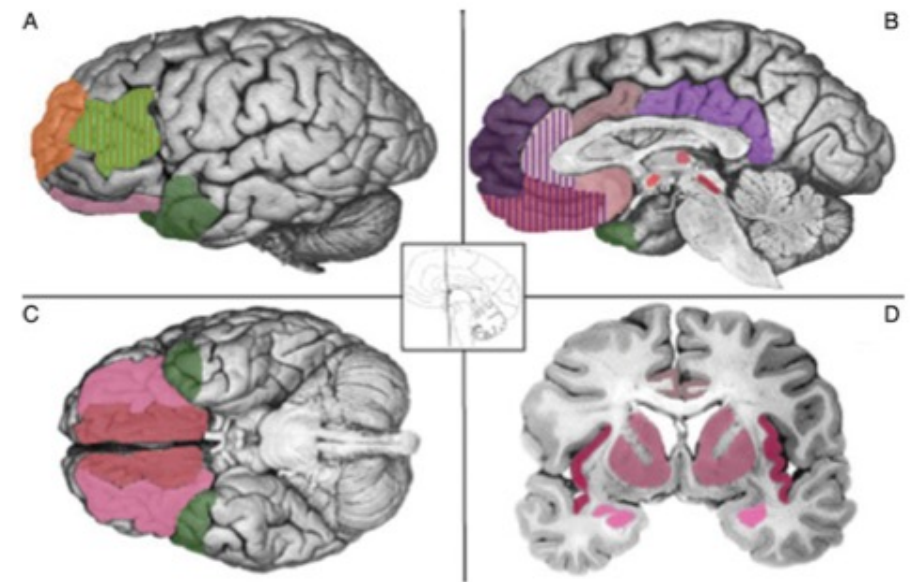
Locationist vs. constructionist account

Locationist Hypotheses of Brain – Emotion Correspondence



Fear = amygdala (yellow); *Disgust* = insula (green);
Anger = OFC (rust); *Sadness* = ACC (blue)

Psychological Constructionist Hypotheses of Brain–Emotion Correspondence



Core affect = amygdala, insula, mOFC, IOFC, ACC, thalamus, hypothalamus, PAG (pink); *Conceptualization* = vmPFC, dmPFC, mTL (e.g., hippocampus), PCC (purple); *Emotion words* = vlPFC, aTL (green); *Executive attention* = dlPFC, vlPFC (orange)

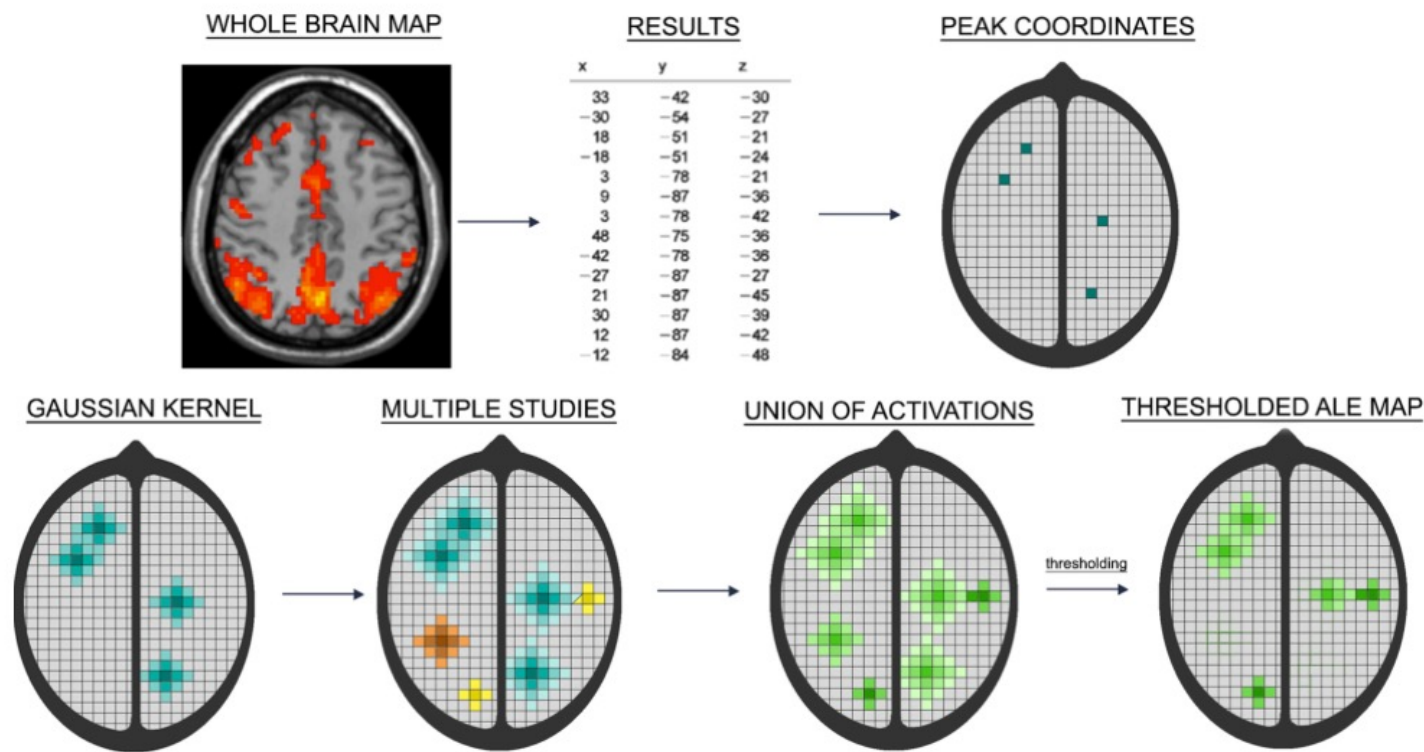
Locationist vs. constructionist account: A meta-analysis

- Analysis of positron emission tomography (PET) or functional magnetic resonance imaging (fMRI) studies targeting the experience or perception of discrete emotions (240 contrasts of anger, sadness, fear, disgust, and happiness from 91 studies published 1993 through the end of 2007)
- Statistically significant peaks of activation within each contrast were included (changes in neural activation relative to neutral baseline)
- Contrasts of emotion experience (i.e., after emotion induction) and perception (e.g., recognition, listening to voices or viewing faces/bodies) were analysed separately (prior meta-analytic evidence found differences in brain regions supporting experience and perception of emotion)
- Studies included healthy adults only

Locationist vs. constructionist account: A meta-analysis

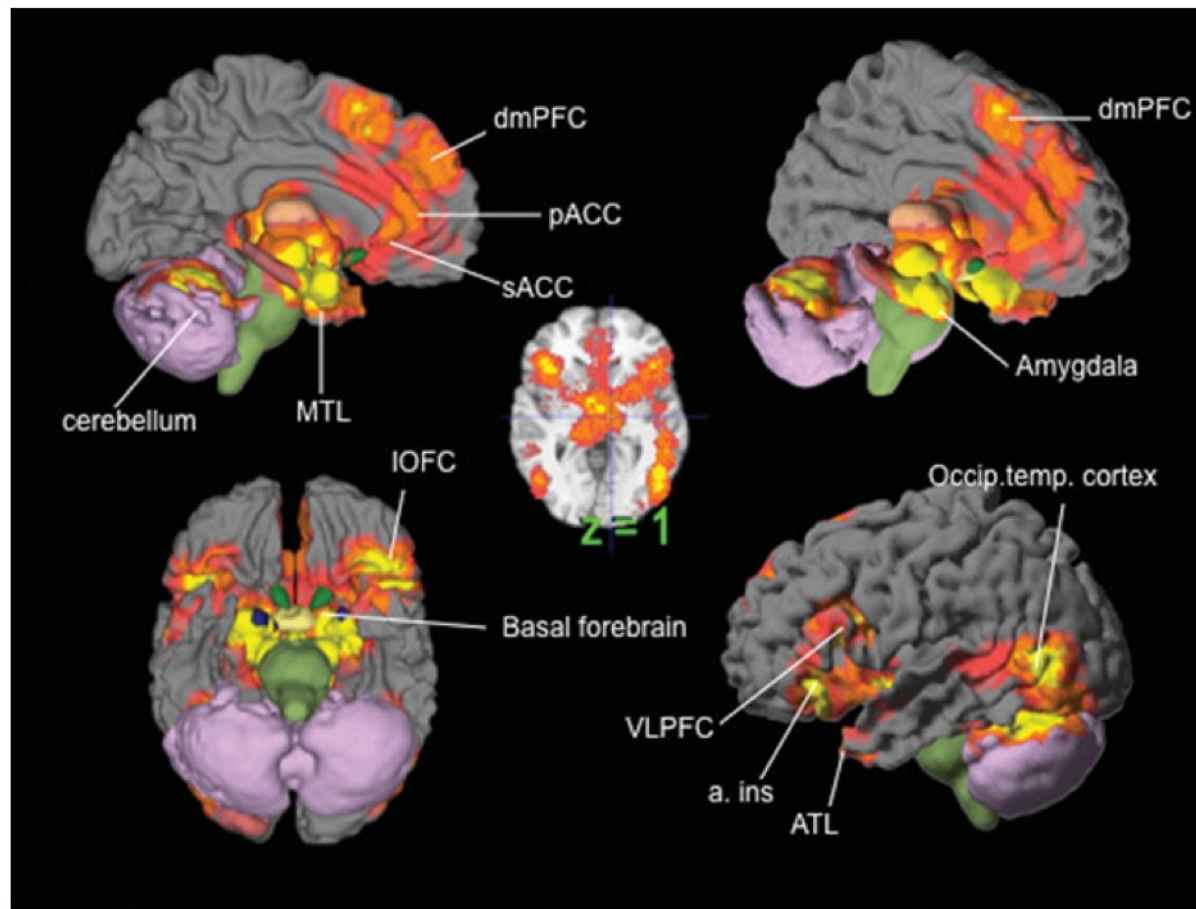
Question: How does one meta-analyze functional neuroimaging studies?

Answer: Activation Likelihood Estimation (ALE)



Locationist vs. constructionist account: A meta-analysis

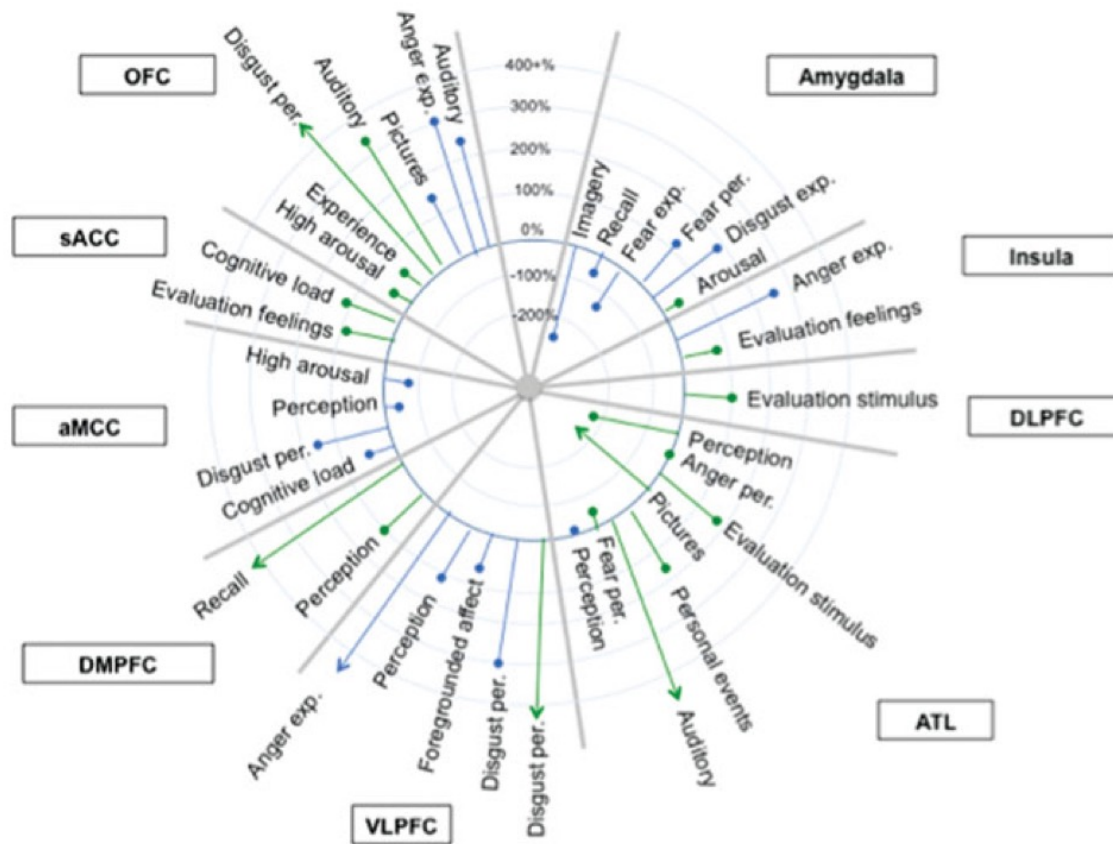
Results (1): Neural reference space for studies of discrete emotions (anger, fear, sadness, disgust, happiness)



This reference space is consistent with the regions hypothesized to support the psychological constructionist perspective on brain-emotion correspondence.

Locationist vs. constructionist account: A meta-analysis

Results (2): No functional specificity in amygdala activation for fear



- More likely to see increased activity in left (blue) amygdala when participants were perceiving instances of fear or experiencing instances of disgust than when perceiving or experiencing any other emotion categories → amygdala responds to salient perceptual stimuli!
- Increases in activity in the right (green) amygdala were likely when participants were experiencing or perceiving instances of any highly arousing emotion category.
- No increase in activity in the left amygdala when participants were focusing on their internal state (i.e., when emotion experience was induced via recall of a personal event and mental imagery) → amygdala responds preferentially to salient exteroceptive (vs. interoceptive) sensations!

Locationist vs. constructionist account: A meta-analysis

Results (3): Consistency versus specificity of brain activation for discrete emotions

Table 3. Summary of Brain Regions Showing Consistent Increases in Activation During Mental States and Methodological Manipulations

	Variable	Area
Mode	Experience of emotion	R. IOFC
	Perception of emotion	DMPFC/dACC R. hippocampus L. VLPFC R. peristriate R. occipitotemporal
Affect	High arousal emotions	R. amygdala R. IOFC PAG
	Unpleasant emotions	L. peristriate
Emotion	Anger experience	L. a. ins.* L. a. ins. L. IOFC L. VLPFC L. ATL
		L. VLPFC*
		R. entorhinal cortex R. DLPFC R. parastriate R. occipitotemporal R. supplementary motor area
	Disgust experience	L. amygdala* R. amygdala L. entorhinal cortex R. IOFC L. occipitotemporal*
		R. IOFC*
	Disgust perception	R. a. ins aMCC L. VLPFC R. VLPFC* R. peristriate R. occipitotemporal
		PAG
	Fear experience	R. peristriate R. parastriate R. occipitotemporal* L. middle temporal
		L. amygdala L. entorhinal cortex R. entorhinal cortex L. hippocampus R. middle temporal
	Happiness experience	L. peristriate
	Sadness experience	L. entorhinal cortex DMPFC R. middle temporal R. putamen

(continued)

Table 3 (Continued)

	Variable	Area
Method Types	Recall	PAG pACC DMPFC/aMCC
	Auditory	R. IOFC R. VLPFC R. ATL
	Imagery	R. occipitotemporal R. middle temporal
	Visual	R. peristriate L. occipitotemporal
	Pictures	L. IOFC R. peristriate R. occipitotemporal
Stimuli	Faces	L. VLPFC
	Films	DMPFC/aMCC L. peristriate R. occipitotemporal R. middle temporal
	Evaluation of feelings	R. a. ins sACC
	Evaluation of stimulus	R. ATL R. DLPFC PAG L. peristriate L. occipitotemporal
	Foregrounded affect	L. VLPFC L. peristriate
Other Psychological Variables	Foregrounded affect	L. VLPFC L. peristriate
	Cognitive load	sACC aMCC

- Many brain regions are consistently activated in the context of emotions.
- There seems to be some functional selectivity.
- The findings do not support functional specificity.

***Lindquist et al. (2012). *The brain basis of emotion: a meta-analytic review*. Behavioral and brain sciences, 35(3), 121-143.

Your turn!



Image created with AI (Bing), October 2024

Which of the core assumptions of psychological constructionist approaches do these meta-analyses not address?

***Discuss with your neighbour(s)
~2 minutes***

Addressing issues of causality and temporality

Lesion studies (e.g., Weiskrantz, 1956):

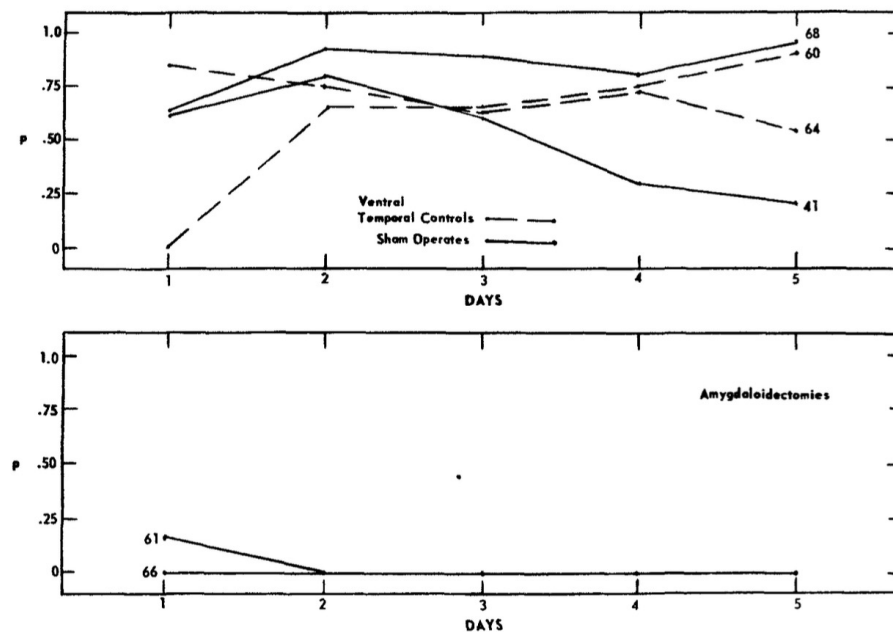


FIG. 5. Proportions of avoidance responses to experimenter.

“[...] marked increase in tameness and a weakening or disappearance of fear responses to previously aversive stimuli by amygdala [lesion] animals”

Addressing issues of causality and temporality

Optogenetic studies (e.g., Tye et al., 2012):



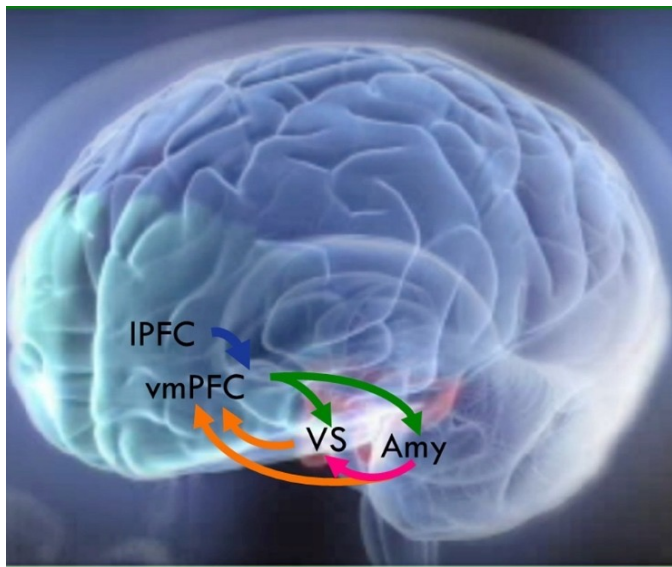
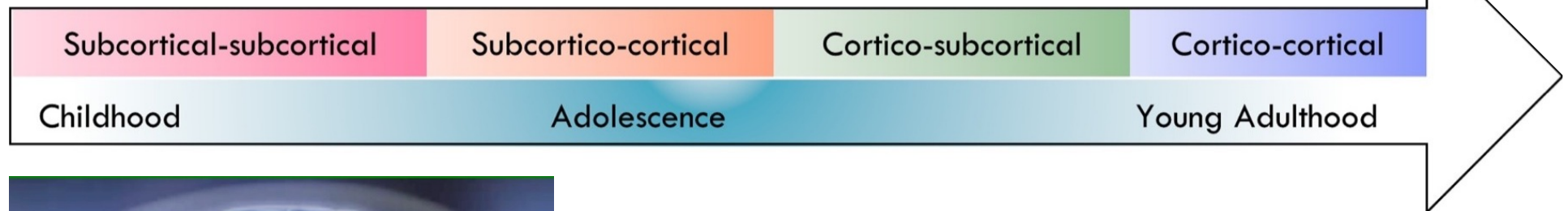
→ Optogenetics

a biological technique which involves the use of light to control cells in living tissue, typically neurons, that have been genetically modified to express light-sensitive ion channels. It is a neuromodulation method that uses a combination of techniques from optics and genetics to control and monitor the activities of individual neurons in living tissue—even within freely-moving animals—and to precisely measure these manipulation effects in real-time.

“[...] temporally precise optogenetic stimulation of basolateral amygdala (BLA) terminals in the central nucleus of the amygdala [...] exerted an acute, reversible anxiolytic effect”

Emotional development

Schematic representation of hierarchical fine-tuning involving different neural structures hypothesized to take place between childhood and adulthood



The ventral striatum (VS) is implicated in learning and prediction of positive outcomes and receives inputs from the basolateral amygdala (Amy). The amygdala is important in learning the emotional significance of cues in the environment, and can facilitate ventral striatum activity through its direct inputs from the basolateral nucleus, leading to motivated action. The ventromedial prefrontal cortex (vmPFC) has been implicated in fear and emotion regulation. This region has dense projections to the (inhibitory) cells in the amygdala and to the ventral striatum (VS) that modulate emotive behaviors. Cortico-cortical connections include pathways between lateral prefrontal cortex (IPFC), implicated in higher cognitive functions, and medial prefrontal regions, involved in emotion and social interactions. These connections may serve as an interface between cognitive and emotional processes.

Current developmental models propose that there is a **hierarchical fine-tuning** that takes place across childhood and adolescence **as a function of biological maturation and experience**; this process represents a potential **mechanism for the observed changes in emotional reactivity and regulation across childhood and adolescence** (e.g., patterns of self-control and risk taking).

Summary

- Meta-analysis of functional neuroimaging studies increases reliability and generalizability by identifying consistent brain activation patterns across diverse studies, while also allowing the examination of moderators such as stimulus features (e.g., emotional valence, complexity) and modalities (e.g., visual, auditory). But, issues of causality may require different approaches (e.g., lesion studies, optogenetics).
- Current models of emotional experience and regulation encompass a number of neural structures with only a few being core emotional regions and most being associated with other aspects of cognitive processing (e.g., memory, decision-making).
- This stance makes clear the somewhat arbitrary distinction between emotional and cognitive processing (cognition is not value free!), and supports a psychological constructionist perspective.
- The hierarchical fine-tuning and maturation of connections between different brain regions throughout childhood, adolescence and early adulthood support the progressive development of cognitive functions, emotional regulation, and behavioral control, enabling more efficient information processing, decision-making, and adaptive responses to complex environmental demands.

Key (mandatory) reading

- Lindquist, K. A., Wager, T. D., Kober, H., Bliss-Moreau, E., & Barrett, L. F. (2012). *The brain basis of emotion: a meta-analytic review*. Behavioral and brain sciences, 35(3), 121-143.
<https://www.cambridge.org/core/services/aop-cambridge-core/content/view/80F95F093305C76BA2C66BBA48D4BC8A/S0140525X11000446a.pdf/the-brain-basis-of-emotion-a-meta-analytic-review.pdf>

BEHAVIORAL AND BRAIN SCIENCES (2012) 35, 121–202

doi:10.1017/S0140525X11000446

The brain basis of emotion: A meta-analytic review

Kristen A. Lindquist

Department of Neurology, Harvard Medical School/Massachusetts General Hospital/Martinos Center for Biomedical Imaging, Charlestown, MA 02129, and Department of Psychology, Harvard University, Cambridge, MA 02138
lindqukr@nmr.mgh.harvard.edu
<http://www.nmr.mgh.harvard.edu/~lindqukr/>

Tor D. Wager

Department of Psychology and Neuroscience, University of Colorado, Boulder, CO 80309
tor.wager@colorado.edu
<http://www.psych.colorado.edu/~tor/>

Hedy Kober

Department of Psychiatry, Yale University School of Medicine, New Haven, CT 06519
hedy.kober@yale.edu
http://medicine.yale.edu/psychiatry/people/hedy_kober.profile

Eliza Bliss-Moreau

California National Primate Research Center, University of California, Davis, CA 95616, and Department of Psychiatry and Behavioral Sciences, University of California, Davis, CA 95616
ebliismoreau@ucdavis.edu
<http://www.elizablissmoreau.com/EBM/home.html>

Lisa Feldman Barrett

Department of Psychology, Northeastern University, Boston, MA 02115, and Departments of Radiology and Psychiatry, Harvard Medical School/Massachusetts General Hospital/Martinos Center for Biomedical Imaging, Charlestown, MA 02129
l.barrett@neu.edu
<http://www.affective-science.org/>

Abstract: Researchers have wondered how the brain creates emotions since the early days of psychological science. With a surge of studies in affective neuroscience in recent decades, scientists are poised to answer this question. In this target article, we present a meta-analytic summary of the neuroimaging literature on human emotion. We compare the locationist approach (i.e., the hypothesis that discrete emotion categories consistently and specifically correspond to distinct brain regions) with the psychological constructionist approach (i.e., the hypothesis that discrete emotion categories are constructed of more general brain networks not specific to those categories) to better understand the brain basis of emotion. We review both locationist and psychological constructionist hypotheses of brain–emotion correspondence and report meta-analytic findings bearing on these hypotheses. Overall, we found little evidence that discrete emotion categories can be consistently and specifically localized to distinct brain regions. Instead, we found evidence that is consistent with a psychological constructionist approach to the mind: A set of interacting brain regions commonly involved in basic psychological operations of both an emotional and non-emotional nature are active during emotion experience and perception across a range of discrete emotion categories.

Keywords: Discrete emotion; emotion experience; emotion perception; meta-analysis; neuroimaging; psychological construction

1. Introduction

William James framed the question of emotion–brain correspondence when he wrote, “of two things concerning the emotions, one must be true. Either separate and special centres, affected to them alone, are their brain-seat, or else they correspond to processes occurring in the motor

and sensory centres already assigned” (James 1890/1998, p. 473). In this target article, we statistically summarize the last 15 years of neuroimaging research on emotion in an attempt to determine which of these alternatives is correct. We examine the utility of two different models of emotion that have each existed since the beginning of psychology.

Additional (optional) reading

